Controlled HiddenBench Imitation

Game definition, feedback metrics, current fluctuations, and the emerging (q, q_c) phase picture

Research report based on the current implementation and completed reasoning-mode experiments

Current status in one paragraph. The experiment has progressed from controller validation to a scalable controlled imitation process with three independent structural parameters: population size N, social interaction size q, and controller sensing size q_c. The soft controller fixes the action-overlap problem that made the deterministic threshold policy difficult to analyse. At $N = 32, q = 1$, increasing q_c mainly improves trajectory reliability: the truth-current variance falls from 29.5 to 8.42 while the precision ratio rises from 0.77 to 2.96. The new $q = 2, q_c = 2$ cell changes the picture much more strongly: the mean truth current falls from 22.85 to 17.26, its variance rises to 176.54, the precision ratio collapses to 0.098, and the target/focal/population actuation CMIs no longer clear their permutation nulls. The current evidence therefore points to a competition between social interaction order and feedback control, with a sharp finite-system regime change that is now worth mapping in the $(q, q_c/N)$ plane.

1 The game: what is being simulated?

1.1 HiddenBench information structure

A task contains a scenario, shared evidence I_s, hidden evidence types $I_u = (u_0, \ldots, u_{C-1})$, a finite answer set $\mathcal{A}$ with $K = |\mathcal{A}|$, and a correct answer o^*. The hidden-profile construction is adversarial in a specific sense: shared evidence favours a decoy, while the correct decision requires pooling the distributed hidden facts. Agents are not told that their information differs.

For large populations, the current experiments use **validated paraphrased replication**: several agents may carry the same evidence *type*, but each receives a semantically equivalent paraphrase rather than a verbatim duplicate. Thus N can be increased without deliberately changing the underlying task-level information structure. The implementation enforces complete evidence-type coverage and balanced allocation across the population.

1.2 Population state and elementary event

Each agent i holds one current option $X_i(t) \in \mathcal{A}$. The macroscopic state is the occupation vector

$$\mathbf{n}(t) = (n_1(t), \ldots, n_K(t)), \qquad n_j(t) = \sum_{i=1}^{N} \mathbf{1}[X_i(t) = j], \qquad \sum_j n_j(t) = N. \tag{1}$$

One elementary event updates at most one agent:

1. choose one focal agent and q distinct ordinary peers;
2. independently, the controller polls q_c agents from the population;
3. the controller selects $U_t \in \{\text{NO\_OP}, \text{ADVOCATE}_Z\}$;
4. under NO_OP, the focal receives all q ordinary social inputs;
5. under ADVOCATE_Z, exactly one of the q social slots is replaced by a controller advocacy message;
6. only the focal agent may change its vote.

Hence the controller is **external**: it has no vote, no private HiddenBench evidence, and never overwrites an agent's choice.

The scaled experiment separates three resources:

$$\boxed{N = \text{population size}, \qquad q = \text{social influence slots}, \qquad q_c = \text{controller sensing size.}} \tag{2}$$

The original protocol is recovered at $q = 1$ and $q_c = 2$.

1.3 Time convention

The process is discrete-time. The event index is the simulation clock; for comparisons across population sizes we use the sweep time

$$\tau = \frac{t}{N}. \tag{3}$$

Thus N elementary focal-update opportunities constitute one population sweep. This is a normalization for exposure across N; it is *not* Gillespie physical time.

2 The feedback controller

The controller implements the factorization

$$\mathbf{n}_t \xrightarrow{S_{q_c}(y|\mathbf{n}_t)} Y_t \xrightarrow{\pi(u|y)} U_t \xrightarrow{P_q(\mathbf{n}_{t+1}|\mathbf{n}_t,u)} \mathbf{n}_{t+1}. \tag{4}$$

2.1 Sensor

Let Z denote the controller target and $p_Z = n_Z/N$. The controller samples q_c agents without replacement and observes

$$s_t = \frac{1}{q_c} \sum_{i \in S_t} \mathbf{1}[X_i(t) = Z]. \tag{5}$$

Conditioned on the population state,

$$q_c s_t \sim \text{Hypergeometric}(N, n_Z, q_c), \tag{6}$$

with

$$\mathbb{E}[s_t] = p_Z, \qquad \text{Var}(s_t) = \frac{p_Z(1-p_Z)}{q_c}\frac{N-q_c}{N-1}. \tag{7}$$

So increasing q_c reduces measurement noise; at $q_c = N$ the target fraction is known exactly.

2.2 Hard and soft policies

The original hard controller is

$$U_t = \text{ADVOCATE}_Z \quad \Longleftrightarrow \quad s_t < \theta. \tag{8}$$

The stochastic controller is

$$\Pr(U_t = \text{ADVOCATE}_Z \mid Y_t) = \sigma(\beta(\theta - s_t)), \qquad \sigma(x) = \frac{1}{1+e^{-x}}. \tag{9}$$

The soft policy was introduced for **identifiability**: at comparable states it gives positive probability to both ADVOCATE_Z and NO_OP. Its purpose is not to inflate mutual information; it is to create the within-state action support needed to estimate actuation channels.

2.3 Reasoning and classical arms

Everything through sensing, controller decision, social-slot replacement, state update, and logging is shared. The only substantive branch is the focal decision rule:

- **reasoning mode:** the focal LLM reads social messages and returns a vote;
- **classical mode:** a provider-free transition kernel samples the destination opinion from explicit weights.

This is the eventual controlled comparison: the LLM implements an implicit transition kernel, while the classical arm provides an explicit stochastic-process reference.

3 Macroscopic observables and information-flow metrics

3.1 Order parameters

For any option Z define the K-state alignment

$$m_Z(t) = \frac{K p_Z(t) - 1}{K - 1}. \tag{10}$$

It equals 1 at consensus on Z, 0 for a uniform population, and $-1/(K-1)$ when nobody supports Z. We use

$$m_{\text{truth}} : Z = o^*, \tag{11}$$
$$m_{\text{ctrl}} : Z = \text{controller target}, \tag{12}$$
$$m_{\text{order}} = \frac{K \max_j p_j - 1}{K - 1}, \tag{13}$$
$$H_{\text{vote}} = -\frac{\sum_j p_j \ln p_j}{\ln K}. \tag{14}$$

When the controller targets the correct answer, $m_{\text{ctrl}} = m_{\text{truth}}$. For the information estimators, the corresponding target headcount is a one-to-one encoding of these alignments, so the mutual information is unchanged by that reparameterization.

3.2 Information metrics: simple and technical readings

Quantity	Plain meaning	Technical definition / caution
Sensing MI	How much the controller learns about the population from its poll.	$I(\mathbf{N}_t; Y_t)$. Also projected onto m_{truth}, m_{ctrl} and m_{order}. Raw values depend on the alphabet and therefore should be compared cautiously when q_c changes.
Population actuation CMI	Does knowing whether the controller advocated help predict what happens next?	$I(U_t; \mathbf{N}_{t+1} \mid \mathbf{N}_t)$. It is unsigned: dependence does not imply movement toward the target.
Target / truth actuation CMI	Does the action change the next target/truth headcount beyond the current headcount?	$I(U_t; m_{t+1} \mid m_t)$, with $m = m_{\text{ctrl}}$ or m_{truth}. Lower dimensional and generally more stable than the full-state CMI.
Focal actuation CMI	Does advocacy influence the one agent actually being updated?	$I(U_t; X^f_{t+1} \mid X^f_t, \mathbf{N}_t)$.
Controller entropy	How much action freedom remains at a fixed state?	$H(U_t \mid S_t)$. It bounds the corresponding CMI: $I(U; S' \mid S) \le H(U \mid S)$.
Information fraction	What fraction of the available action information predicts the next state?	$\rho_S = I(U; S' \mid S)/H(U \mid S)$. A normalization diagnostic, **not** a thermodynamic efficiency.
Signed actuation	Does advocacy move an order parameter up or down relative to no-op?	State-adjusted $\mathbb{E}[\Delta m \mid \text{ADVOCATE}_Z, S] - \mathbb{E}[\Delta m \mid \text{NO\_OP}, S]$ over states with both actions.
Action overlap	Do we actually observe both actions at comparable states?	Fraction of events in conditioning states containing both ADVOCATE_Z and NO_OP. This is the key identifiability diagnostic.

3.3 Truth current and fluctuations

At each event,

$$j_t^{\text{truth}} = \begin{cases} +1, & X_t^f \neq o^*,\ X_{t+1}^f = o^*, \\ -1, & X_t^f = o^*,\ X_{t+1}^f \neq o^*, \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

The episode current is

$$J_{\text{truth}} = \sum_{t=0}^{T-1} j_t^{\text{truth}} = N_{\to\text{truth}} - N_{\leftarrow\text{truth}}. \tag{16}$$

Because only one focal opinion changes per event,

$$\boxed{J_{\mathrm{truth}} = n_{\mathrm{truth}}(T) - n_{\mathrm{truth}}(0).} \tag{17}$$

Plain meaning. J_{truth} is the net gain in truth-supporting agents over the episode. **Technical meaning.** It telescopes, so it does not measure path volatility. Two trajectories with the same start and end truth headcount have the same J_{truth} even if one contains many back-and-forth switches.

Across equal-horizon episodes, the current-precision statistic is

$$F_{\mathrm{truth}} = \frac{|\langle J_{\mathrm{truth}} \rangle|}{\mathrm{Var}(J_{\mathrm{truth}})}. \tag{18}$$

A larger value means the net truth-directed current is more reproducible relative to its across-episode fluctuations. At present this is an empirical current precision ratio; **no thermodynamic uncertainty-relation claim is made for the LLM system**.

4 Result I: validating the soft controller

The first large comparison asked whether the tiny actuation CMIs obtained with the hard threshold policy were genuine or simply caused by a nearly deterministic action at fixed state.

Table 1: Controller-validation experiment. The soft policy creates action overlap and makes one-step actuation statistically identifiable.

Statistic	Threshold	Soft	Interpretation
$H(U)$	0.3879	0.8445	More action variability under the soft policy.
$H(U \mid \mathbf{N})$	0.1442	0.6924	About 4.8× more action uncertainty is available at fixed population state.
Events in dual-action $\mathbf{N}_t$ states	17.5%	99.8%	The soft dataset is almost entirely supported by genuine within-state action comparisons.
$I(U; \mathbf{N}' \mid \mathbf{N})$	0.0121	0.0285	Raw population actuation information increases.
Permutation-null mean	0.0110	0.0098	Hard-control population CMI is near null; soft-control CMI is clearly above it.
$I(U; m'_{\text{truth}} \mid m_{\text{truth}})$	0.0034	0.0172	Target/truth actuation becomes clearly measurable.
Target-CMI null mean	0.0046–0.0048	0.0027–0.0028	Under hard control the target channel is not separated from null; under soft control it is.
Population information fraction	0.0840	0.0412	Soft control creates more raw actuation information, but it is still a small fraction of the available action budget.
Truth information fraction	0.0234	0.0248	About 2.5% of action information at fixed truth state predicts the next truth state in both cases.

Plain conclusion. The soft controller solved the statistical problem. We now observe both advocacy and no-op in the same states, so the actuation quantities can be interpreted. Advocacy does affect the next-step transition distribution, but the immediate effect is modest relative to the controller's available action entropy.

Technical conclusion. The earlier small CMI was partly an action-support problem, but not only that. Once $H(U \mid S)$ and overlap are restored, the CMI rises above the permutation null yet remains far below its entropy ceiling. Therefore the system exhibits detectable but weak one-step actuation in this regime.

The signed soft target response in this first validation run was positive but uncertain (0.0057, bootstrap interval approximately $[-0.002, 0.014]$). Thus the strongest supported statement was *predictive influence*, not yet a clean positive one-step causal drift toward the target.

5 Result II: increasing controller sensing at $N = 32$, $q = 1$

The new sweep holds the population and social interaction fixed and varies only the controller sensor size:

$$N = 32, \qquad q = 1, \qquad q_c \in \{2, 8, 32\}, \tag{19}$$

using reasoning dynamics and the soft-target controller.

Table 2: Current q_c sweep. CMI values are unsmoothed plug-in estimates; current statistics are across equal-horizon episodes.

q_c	Episodes	$I(\mathbf{N};Y)$	$H(U \mid \mathbf{N})$	$I(U;m'_{\text{truth}} \mid m_{\text{truth}})$	$\langle J_{\text{truth}}\rangle$	$\text{Var}(J_{\text{truth}})$	F_{truth}
2	48	0.4832	0.8102	0.0321	22.854	29.532	0.774
8	46	1.4819	0.7530	0.0251	23.652	22.721	1.041
32	48	6.3362	0.7353	0.0235	24.958	8.424	2.963

The target CMI is well above its permutation null in every cell: approximately 0.0321 vs 0.0030 at $q_c = 2$, 0.0251 vs 0.0032 at $q_c = 8$, and 0.0235 vs 0.0030 at $q_c = 32$. Action overlap remains excellent: about 99.6–99.7% of events lie in full-population states supporting both controller actions, and essentially 100% of events have dual-action support in the truth/target conditioning space.

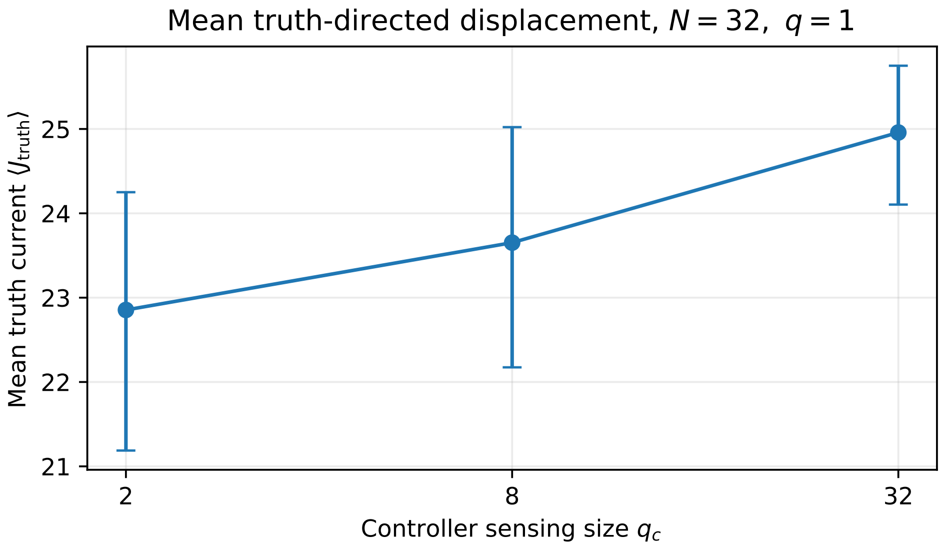

Figure 1: Increasing q_c produces only a moderate increase in the mean net truth current. Error bars are the reported bootstrap intervals.

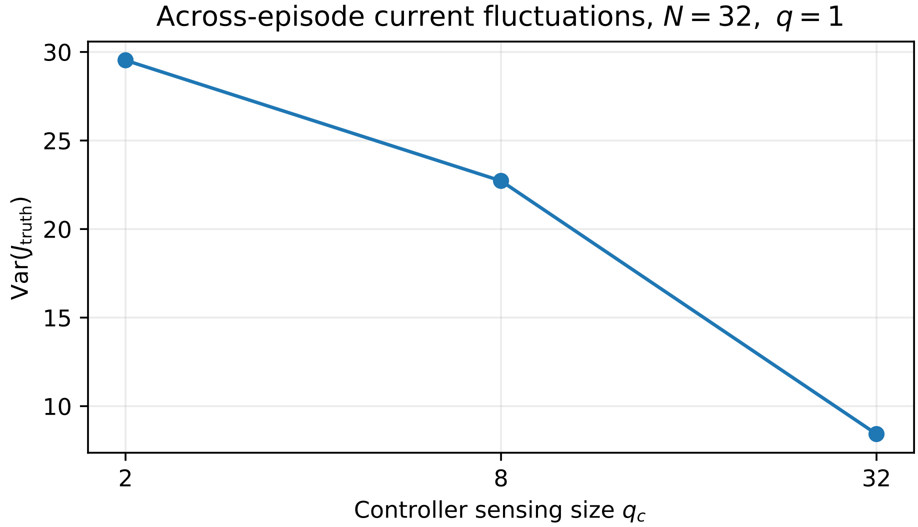

Figure 2: The strongest effect of increasing q_c is a large reduction in across-episode fluctuations of the truth current.

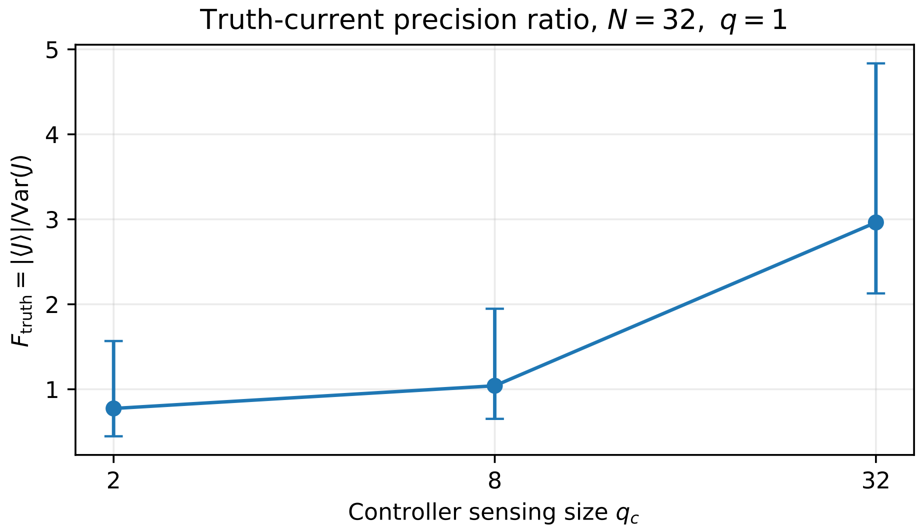

Figure 3: The current precision ratio rises almost fourfold from $q_c = 2$ to $q_c = 32$. Error bars are the reported bootstrap intervals.

5.1 What this means in plain terms

Better sensing does *not* make each advocacy message dramatically more powerful. Instead, it makes the overall episode outcome much more reliable. The average net gain toward truth increases by only about 9% from $q_c = 2$ to $q_c = 32$, but the variance of that gain falls by about 71%. The controller with full-population sensing reaches roughly the same direction of travel with much less episode-to-episode uncertainty.

5.2 What this means technically

The results separate **measurement quality** from **local actuation strength**. As q_c grows, sensing becomes more informative and the controller action becomes slightly more predictable given the state, as seen in the decline of $H(U \mid \mathbf{N})$ from 0.810 to 0.735 bits. Yet the one-step truth actuation CMI declines slightly rather than increasing. Thus the principal benefit of better sensing is not a larger instantaneous action-to-next-state information channel; it appears at the trajectory level as reduced fluctuations and improved precision.

A useful derived normalization makes this even clearer:

$$\rho_{\mathrm{truth}} = \frac{I(U_t; m_{\mathrm{truth},t+1} \mid m_{\mathrm{truth},t})}{H(U_t \mid m_{\mathrm{truth},t})} \approx 0.039,\ 0.033,\ 0.032 \tag{20}$$

for $q_c = 2, 8, 32$ respectively. The immediate fraction of available action information expressed in the next truth state remains only a few percent.

5.3 Important estimator caveat

Raw sensing MI should **not** be treated as an order parameter across different q_c. The sensor alphabet changes with q_c, and the $q_c = 32$ full-state estimate is estimator-sensitive: the unsmoothed value is 6.336 bits whereas the Jeffreys estimate is about 1.895 bits. Because the sensor kernel is known analytically, future phase diagrams should rely more heavily on direct sensor error, hypergeometric variance, or carefully normalized/projected information measures than on raw $I(\mathbf{N}; Y)$ alone.

6 Result III: increasing social interaction from $q = 1$ to $q = 2$

The first new point away from the $q = 1$ line keeps both population size and controller sensing fixed,

$$N = 32, \qquad q_c = 2, \qquad q \in \{1, 2\}, \tag{21}$$

so the change is attributable to the social interaction structure of the reasoning dynamics. Under advocacy the controller still replaces exactly one influence slot; therefore its immediate share of the focal influence set changes from 1 at $q = 1$ to 1/2 at $q = 2$.

Table 3: Matched comparison at $N = 32$ and $q_c = 2$. "Null" denotes the within-episode action-permutation null mean.

Quantity	**q = 1**	**q = 2**	**Reading**
Episodes	48	47	Comparable replication.
$\langle J_{\mathrm{truth}} \rangle$	22.854	17.255	Net truth gain drops by about 24%.
$\mathrm{Var}(J_{\mathrm{truth}})$	29.532	176.542	Across-episode current fluctuations increase by about 6×.
F_{truth}	0.7739	0.0977	Current precision falls by about 87%.
Population actuation CMI	0.0765	0.0384	One-step population dependence is roughly halved.
Population CMI null	0.0505	0.0421	At $q = 2$ the direct estimate is no longer above null.
Truth/target actuation CMI	0.0321	0.0043	Truth-directed one-step channel collapses.
Truth/target CMI null	0.0030	0.0052	At $q = 2$ the direct estimate is at/below null.
Focal actuation CMI	0.0766	0.0309	The focal-response channel also weakens strongly.
Focal CMI null	0.0476	0.0339	Again no clear separation from null at $q = 2$.
$H(U \mid \mathbf{N})$	0.8102	0.7282	Action-information budget remains large.
$H(U \mid m_{\mathrm{truth}})$	0.8225	0.7399	The weak CMI is not caused by a degenerate controller.
Dual-action event fraction in $\mathbf{N}_t$	99.7%	99.3%	Within-state action support remains excellent.

6.1 Plain interpretation

The change from one peer slot to two makes the dynamics much less reliable. The population still moves toward the correct answer on average, but it does so less strongly and with far larger episode-to-episode variability. At the same time, whether the controller advocates or stays silent no longer tells us much about the immediate next truth state once the present state is known.

This is qualitatively different from increasing q_c. Better sensing at $q = 1$ mainly reduced fluctuations while preserving a detectable local actuation channel. Increasing q from 1 to 2 instead weakens the local actuation signal and simultaneously amplifies trajectory-level fluctuations.

6.2 Technical interpretation

The $q = 2$ result is not an identifiability failure. The controller has substantial conditional entropy and essentially complete action overlap, yet

$$I(U_t; m_{\mathrm{truth},t+1} \mid m_{\mathrm{truth},t}) = 0.0043 \text{ bits} \tag{22}$$

compared with a permutation-null mean of 0.0052 bits. Likewise, the population and focal CMIs do not clearly clear their nulls. A useful normalized diagnostic is

$$\rho_{\text{truth}} = \frac{I(U_t; m_{\text{truth},t+1} \mid m_{\text{truth},t})}{H(U_t \mid m_{\text{truth},t})}, \tag{23}$$

which falls from about 0.039 at $q = 1, q_c = 2$ to about 0.0058 at $q = 2, q_c = 2$. In other words, less than one percent of the available action uncertainty is expressed in the immediate next truth state in the $q = 2$ cell.

The trajectory statistics shift even more strongly. The mean-current bootstrap intervals do not overlap: $[21.19, 24.25]$ for $q = 1$ versus $[13.47, 20.74]$ for $q = 2$. The precision intervals are also widely separated: approximately $[0.447, 1.567]$ versus $[0.066, 0.162]$. These observations support a real finite-system regime change in convergence reliability, even though they do not establish a thermodynamic phase transition.

6.3 Why the result is physically interesting

At fixed q_c, increasing q changes the balance between ordinary social influence and external feedback. The focal receives more peer context, while one controller intervention still occupies only one slot. Mechanically, the controller fraction of the immediate influence set scales as $1/q$. Semantically, a larger peer set may also create stronger or more heterogeneous social evidence. The present experiment intentionally combines these effects; separating them later by holding the controller-slot fraction fixed would be a useful ablation.

A second important observation is that the sensing MI increases at $q = 2$ even though q_c is unchanged. This should not be read as a better sensor. The sensor kernel is the same; changing q changes the distribution of population states visited by the dynamics, and mutual information depends on that state distribution. This is another reason not to use raw sensing MI as the phase order parameter.

7 What can already be concluded?

1. **The soft controller is experimentally validated.** It provides nearly complete action overlap while preserving the same feedback semantics.
2. **At $q = 1$, one-step actuation is detectable but modest.** Target, focal, and population CMI clear permutation nulls, while remaining small relative to $H(U \mid S)$.
3. **At $q = 1$, increasing q_c mainly improves trajectory reliability.** The truth-current variance falls sharply and F_{truth} rises, even though the one-step target CMI does not increase.
4. **Changing q from 1 to 2 has a much larger qualitative effect than changing q_c along the $q = 1$ line.** The truth current becomes substantially less precise and the one-step actuation channels no longer clearly separate from their nulls.
5. **The current evidence points to competing control and social-interaction regimes.** The natural next question is whether larger q_c can rescue the $q = 2$ regime, defining a crossover line $q_c^\star(q)$, or whether the one-slot controller remains weak even with nearly perfect sensing.

What is *not* yet supported is equally important. We do not yet have evidence for a thermodynamic phase transition, a thermodynamic efficiency, or a TUR for the LLM dynamics. We have a sharp finite-system regime change and an increasingly well-defined program for testing whether it sharpens with N.

8 Toward a phase diagram

The current parameterization is now well suited for a phase-diagram program. A natural empirical control plane is

$$\boxed{\text{horizontal: } q \quad \text{and} \quad \text{vertical: } r_c = \frac{q_c}{N}.} \tag{24}$$

The four currently populated points already suggest why this plane is useful. Along $q = 1$, increasing q_c/N moves the system toward higher truth-current precision. Moving from $q = 1$ to $q = 2$ at the lowest sensing fraction instead produces a strong loss of precision and of detectable one-step actuation. The missing $q = 2, q_c \in \{8, 32\}$ cells are therefore especially diagnostic: they test whether additional sensing shifts the system back toward the high-precision regime or whether social interaction order dominates the one-slot controller. Here q controls the number of social influence slots and r_c controls the fraction of the population sensed by the controller. Population size N then becomes the finite-size dimension.

8.1 Recommended observables on the $(q, q_c/N)$ plane

Panel / observable	Why it matters	Interpretation
$\langle m_{\text{truth}}(T)\rangle$ or $\langle J_{\text{truth}}\rangle/N$	Primary truth-directed order parameter.	Ordered truth phase vs weak/incorrect truth alignment.
$\chi_{\text{truth}} = N\,\mathrm{Var}[m_{\text{truth}}(T)]$ (proposed)	Susceptibility-like finite-size fluctuation measure.	A sharpening peak with N would be evidence for a transition rather than a smooth crossover.
$\mathrm{Var}(J_{\text{truth}})/N$	Trajectory-scale outcome fluctuations.	Separates reliable from noisy control regimes.
F_{truth}	Current precision.	Highlights the high-precision regime already visible in the q_c sweep.
$I(U; m'_{\text{truth}} \mid m_{\text{truth}})$ and ρ_{truth}	Local actuation channel.	Shows whether the phase-level response is accompanied by stronger one-step information transfer.
H_{vote} and m_{order}	Consensus independent of correctness.	Distinguishes truth alignment from generic herding / premature consensus.

8.2 Why q is especially interesting

In the reasoning experiment, changing q has two linked effects. First, the focal receives a larger social context. Second, the controller still occupies exactly one slot when it acts, so its immediate share of the influence set is roughly $1/q$. Thus q simultaneously controls social interaction order and mechanically dilutes the intervention. This is scientifically meaningful, but it means a q-dependence should be interpreted as a competition between richer peer pressure and smaller per-event controller leverage.

A later control could separate these effects by allowing b controller slots and holding b/q fixed, but the current one-slot protocol is cleaner as the first phase-diagram definition.

8.3 When would we call it a phase transition?

A heatmap alone is not enough. The evidence should become stronger with increasing N. A convincing sequence would be:

1. estimate $\langle m_{\text{truth}}\rangle$, χ_{truth}, current variance, and precision on a dense (q, q_c) grid;
2. repeat for $N \in \{8, 16, 32\}$ (and larger if annotation capacity permits), keeping the number of sweeps fixed;
3. identify ridges or steep slopes that sharpen with N;
4. perform finite-size scaling around the candidate boundary.

Until this sharpening is observed, the correct language is *regime change* or *crossover*.

9 High-value next analyses and experiments

9.1 1. Add activity next to net current

Because J_{truth} telescopes, retain and analyse the path activity

$$A_{\text{truth}} = N_{\to\text{truth}} + N_{\leftarrow\text{truth}}. \tag{25}$$

Then J_{truth} measures directed drift, while A_{truth} measures total switching traffic. This pairing is much closer to the current/activity decomposition used in stochastic thermodynamics and prevents a quiet trajectory from being confused with a highly volatile trajectory that ends at the same state.

9.2 2. Complete the $q = 2$ sensing line, then expand the grid

The immediate empirical priority is now $q = 2$ with $q_c \in \{8, 32\}$ at $N = 32$. This distinguishes two hypotheses. If additional sensing restores truth-current precision and actuation CMI, the data support a controller-resource boundary $q_c^\star(q)$. If full-population sensing does not rescue $q = 2$, then the social interaction structure is suppressing one-slot feedback at the actuation stage rather than merely through sensing noise. After this diagnostic line is complete, expand to a minimal grid such as $q \in \{1, 2, 4, 8\}$ and $q_c/N \in \{1/16, 1/4, 1\}$, subject to cost and integer constraints.

9.3 3. Repeat across N with matched sweep time

Use $T = SN$ so that every population receives the same number of update opportunities per agent. Report intensive quantities (J/N, final truth fraction, m) and fluctuation quantities with an explicit N normalization.

9.4 4. Run reasoning OFF as the controlled reference

The main scientific comparison remains

$$\begin{array}{r|cc} & \text{no control} & \text{feedback} \\ \hline \text{reasoning ON} & A & B \\ \text{reasoning OFF} & C & D \end{array} \tag{26}$$

with matched task, initial state, N, q, q_c, policy, horizon, and seeds where possible. This asks whether semantic reasoning changes controllability and fluctuations relative to a tractable imitation process.

Important implementation caveat. The current classical kernel does not yet consume the sampled q-peer social context: q is logged but discarded by the classical transition rule. Therefore a genuine classical q-phase diagram requires extending the classical kernel before interpreting q as the nonlinear voter interaction parameter. The present reasoning-mode q axis is real; the corresponding classical theoretical axis is not yet implemented.

9.5 5. Wrong-target control

Choose a controller target $Z \neq o^*$. Then m_{ctrl} and m_{truth} separate. This gives a clean adversarial-control experiment: a controller can succeed in raising its own target while simultaneously reducing truth. This is one of the most direct tests of whether reasoning resists or amplifies social persuasion.

9.6 6. Improve sensing diagnostics before using them in phase plots

Since the sensor model is explicitly hypergeometric, use direct quantities such as mean absolute sensor error, sensor MSE, or the known conditional variance alongside MI. If an information normalization is desired, use a clearly defined normalized MI and verify estimator stability across N and q_c.

9.7 7. Connect the classical model to stochastic thermodynamics

The eventual theory target is an explicit controlled stochastic process with sensor kernel S_{q_c}, policy π, and transition kernel P_q. Once the classical q dependence and an appropriate time/reversibility convention are fixed, one can compute stationary laws, affinities, entropy production, currents, activities, and test information-feedback bounds. Only then should the LLM current-precision results be compared against a genuine TUR-like prediction.

10 Current working scientific picture

The evidence so far suggests a useful decomposition:

$$\boxed{\text{sensing quality} \longrightarrow \text{policy decisions} \longrightarrow \text{local actuation} \longrightarrow \text{trajectory reliability.}} \quad (27)$$

The first controller-validation experiment established that the local actuation channel can be measured once action overlap is restored. The $q = 1$ sensing sweep then showed that better sensing can improve *trajectory reliability* much more strongly than it improves one-step actuation. The new $q = 2$ cell adds the complementary result: increasing social interaction order can simultaneously destroy one-step actuation detectability and amplify macroscopic current fluctuations, even though the controller remains stochastic and well supported. This is precisely why the combination of information-flow metrics, order parameters, and current fluctuations is more informative than any one metric alone.

The most promising near-term scientific target is therefore not "maximize the CMI," but to map where the population is truth-aligned, where it is ordered but wrong, where control is locally detectable, and where the truth current becomes precise. A finite-size $(q, q_c/N)$ study can turn these into distinct regimes and determine whether any boundary sharpens into a genuine transition.

A Raw numerical summary

A.1 Controller validation

Statistic	Threshold	Soft target
$H(U)$	0.3879	0.8445
$H(U \mid \mathbf{N})$	0.1442	0.6924
Dual-action event fraction in $\mathbf{N}_t$	0.175	0.998
Population CMI	0.0121	0.0285
Population CMI null mean	0.0110	0.0098
Truth/target CMI	0.0034	0.0172
Truth/target CMI null mean	≈ 0.0047	≈ 0.0028
Focal CMI	0.0129	0.0292
Population information fraction	0.0840	0.0412
Truth information fraction	0.0234	0.0248
Soft signed truth response	–	0.0057 (CI crosses zero)

A.2 $N = 32$, $q = 1$, soft-target q_c sweep

Statistic	$q_c = 2$	$q_c = 8$	$q_c = 32$
Episodes	48	46	48
Events	15360	14720	15360
Sensing MI	0.4832	1.4819	6.3362
Population CMI	0.0765	0.0577	0.0581
Population null mean	0.0505	0.0401	0.0392
Truth/target CMI	0.0321	0.0251	0.0235
Truth/target null mean	0.0030	0.0032	0.0030
$H(U \mid \mathbf{N})$	0.8102	0.7530	0.7353
$H(U \mid m_{\text{truth}})$	0.8225	0.7615	0.7451
Dual-action event fraction in $\mathbf{N}_t$	0.997	0.996	0.997
$\langle J_{\text{truth}} \rangle$	22.8542	23.6522	24.9583
$\mathrm{Var}(J_{\text{truth}})$	29.5315	22.7208	8.42376
F_{truth}	0.7739	1.0410	2.9629

A.3 $N = 32$, $q = 2$, $q_c = 2$, soft target

Statistic	Value
Episodes / events	47 / 15040
Sensing MI	0.7640 (null 0.2172)
Population CMI	0.0384 (null 0.0421)
Truth/target CMI	0.0043 (null 0.0052)
Focal CMI	0.0309 (null 0.0339)
$H(U \mid \mathbf{N})$	0.7282
$H(U \mid m_{\text{truth}})$	0.7399
Dual-action event fraction in $\mathbf{N}_t$	0.993
$\langle J_{\text{truth}} \rangle$	17.2553, CI $[13.4676, 20.7447]$
$\text{Var}(J_{\text{truth}})$	176.542
F_{truth}	0.09774, CI $[0.06609, 0.16229]$

B Source basis and interpretation boundaries

This report is grounded in the current `imitation_game_mechanics.md`, the completed threshold/soft controller analysis reports, and the $N = 32$, $q = 1$, $q_c \in \{2, 8, 32\}$ reports, and the new matched $N = 32$, $q = 2$, $q_c = 2$ information and truth-current reports supplied with this analysis. The mechanics document itself records several limitations that matter for future theory: paraphrase pools were self-verified rather than cross-model verified; disclosure detection is a lower-bound lexical detector; the classical realized chain uses a focal-conditioned embedded jump convention rather than a Gillespie clock; and the current classical kernel does not yet implement a nonlinear q-voter transition law.

All claims labelled as “phase diagram”, “susceptibility”, “activity”, finite-size scaling, or classical q-extension above are **proposed next analyses**, not quantities already established by the current files. Likewise, F_{truth} is treated only as an empirical precision ratio and not as a thermodynamic uncertainty relation for LLM dynamics.